Customer Churn Prediction Project Report

Project Overview

This project predicts customer churn (Exited) using machine learning. The workflow includes data loading, cleaning, exploratory data analysis (EDA), colorful visualizations, feature engineering, model training, prediction, and evaluation.

Step 1: Data Loading

Loaded the Churn_Modelling.csv dataset using Pandas and reviewed the first records.

Step 2: Dataset Understanding

Checked dataset structure using df.info() and reviewed rows and columns.

Step 3: Data Cleaning

Verified missing values, checked duplicates, and removed unnecessary columns: RowNumber, CustomerId, and Surname.

Step 4: Exploratory Data Analysis (EDA)

Analyzed churn distribution, churn percentage, average balance by geography, average age by gender, and churn count by gender.

Step 5: Colorful Visualizations

Created:
• Pie Chart – Customer Churn Distribution
• Histogram – Age Distribution
• Count Plot – Gender vs Churn
• Correlation Heatmap – Feature Relationships

Recommended GitHub visuals:
🔵 Sky Blue Pie Chart
🟣 Purple Age Histogram
🟢 Green/Orange Gender Count Plot
🌈 Coolwarm Correlation Heatmap

Step 6: Feature Engineering

Applied one-hot encoding to Geography and Gender columns using pd.get_dummies().

Step 7: Train-Test Split

Separated features and target variable and split the data into training and testing datasets.

Step 8: Machine Learning Model

Trained a Logistic Regression model to predict customer churn.

Step 9: Predictions

Generated churn predictions on the test dataset.

Step 10: Model Evaluation

Evaluated model performance using Accuracy Score, Confusion Matrix, and Classification Report.

GitHub README Structure

1. Business Problem
2. Dataset Description
3. Data Cleaning
4. EDA Insights
5. Visualizations
6. Feature Engineering
7. Model Building
8. Evaluation Metrics
9. Results & Conclusion
10. Future Improvements

Conclusion

The project successfully demonstrates a complete customer churn prediction workflow from raw data to machine learning evaluation.